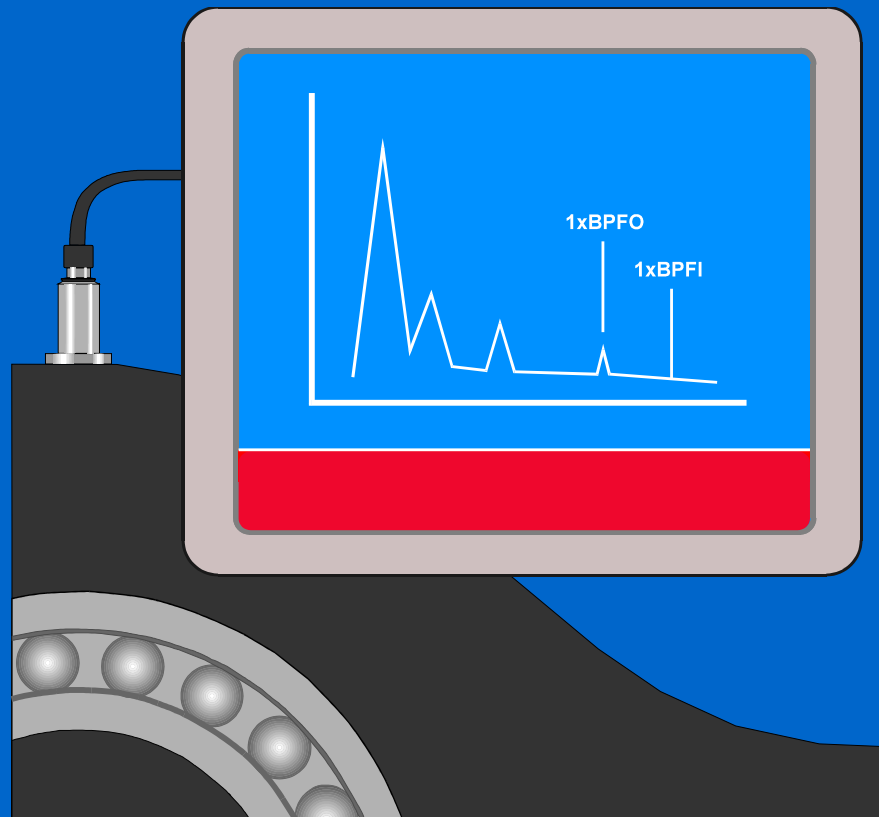


Vibration Analysis

Conducted by:

Anis Lutfi

Condition Monitoring, SKF Reliability
Systems PT. Skefindo Primatama -
INDONESIA



1

SKF PLATFORM

SKF GROUP

FACTORY UNIT

- 80 Factories
- 1 in Indonesia
- 45,000 employees
- ISO 14001

SALES UNIT

- 130 COUNTRY
- 7,000 DISTRIBUTOR

SKF SALES ASIA PACIFIC

SKF SINGAPORE

SKF VIETNAM

SKF THAILAND

SKF TAIWAN

SKF KOREA

SKF PHILIPINA

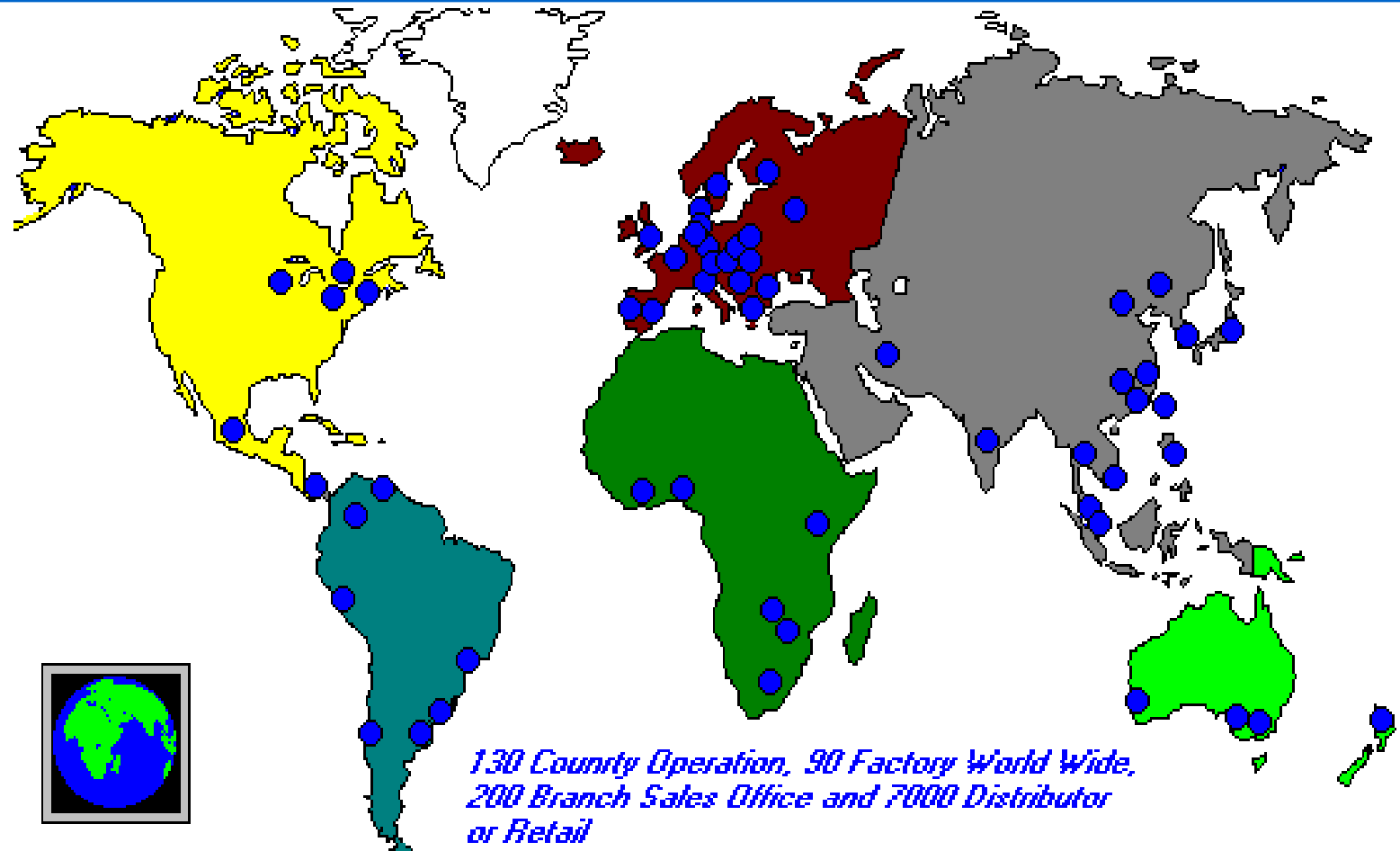
SKF MALAYSIA

SKF INDONESIA (PT.SKEFINDO) etc.

RELIABILITY SYSTEMS INDONESIA

SKF

SKF Network



PT. SKF INDONESIA

SKF

2

MAINTENANCE PHILOSOPHY

The evolution continues...

Not to fix failed machinery, but to prevent machinery failing.

- To maximize productivity.
 - To ensure machinery operates satisfactorily and continuously for its rated life - or longer.
 - To provide the highest machine availability, with minimum downtime due to maintenance and service.
 - To continuously improve the production process.
- To optimize machine performance.
 - Smoother, efficient running machinery costs less to operate, and produces a higher quality product.
- To provide a safe environment.



Slide 7

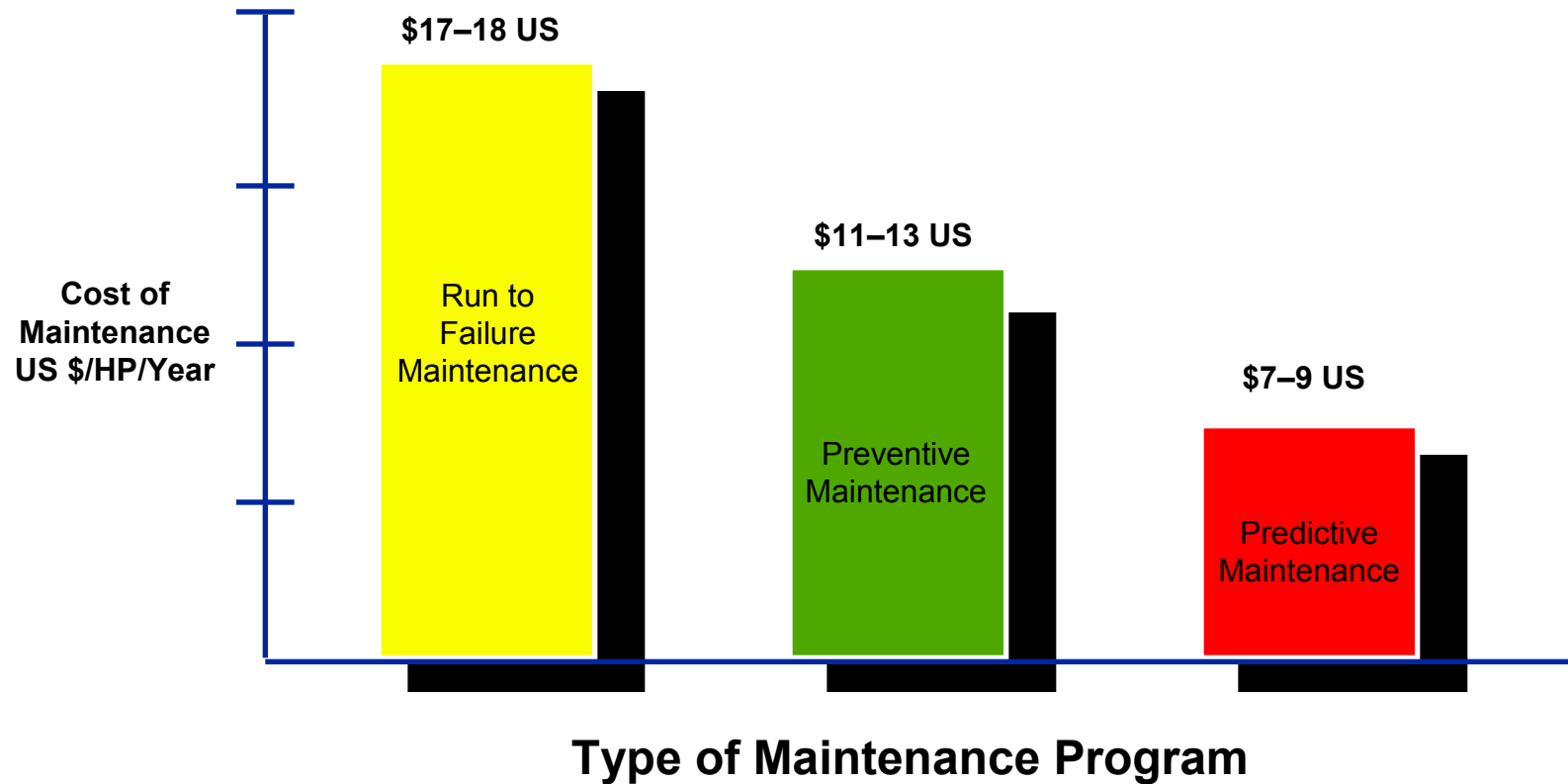
15-07-08 ©SKF Slide 7

SKF Reliability Systems - Indonesia
Jaka Rusdianto

BERKUALITAS DAN BERKUALITAS

SKF

Cost of Maintenance



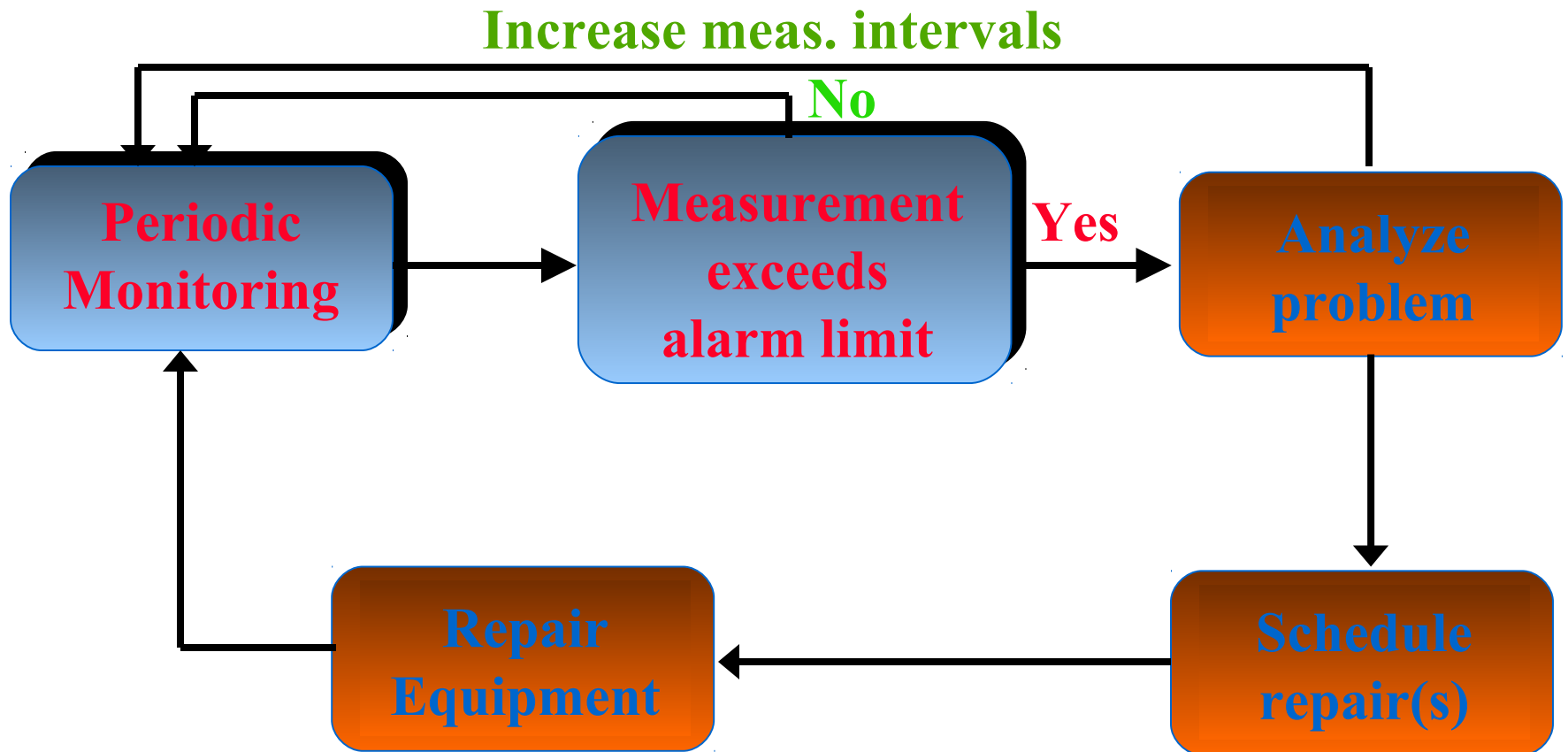
Customer : Cevron Oil USA

PT. CEBRON OIL USA

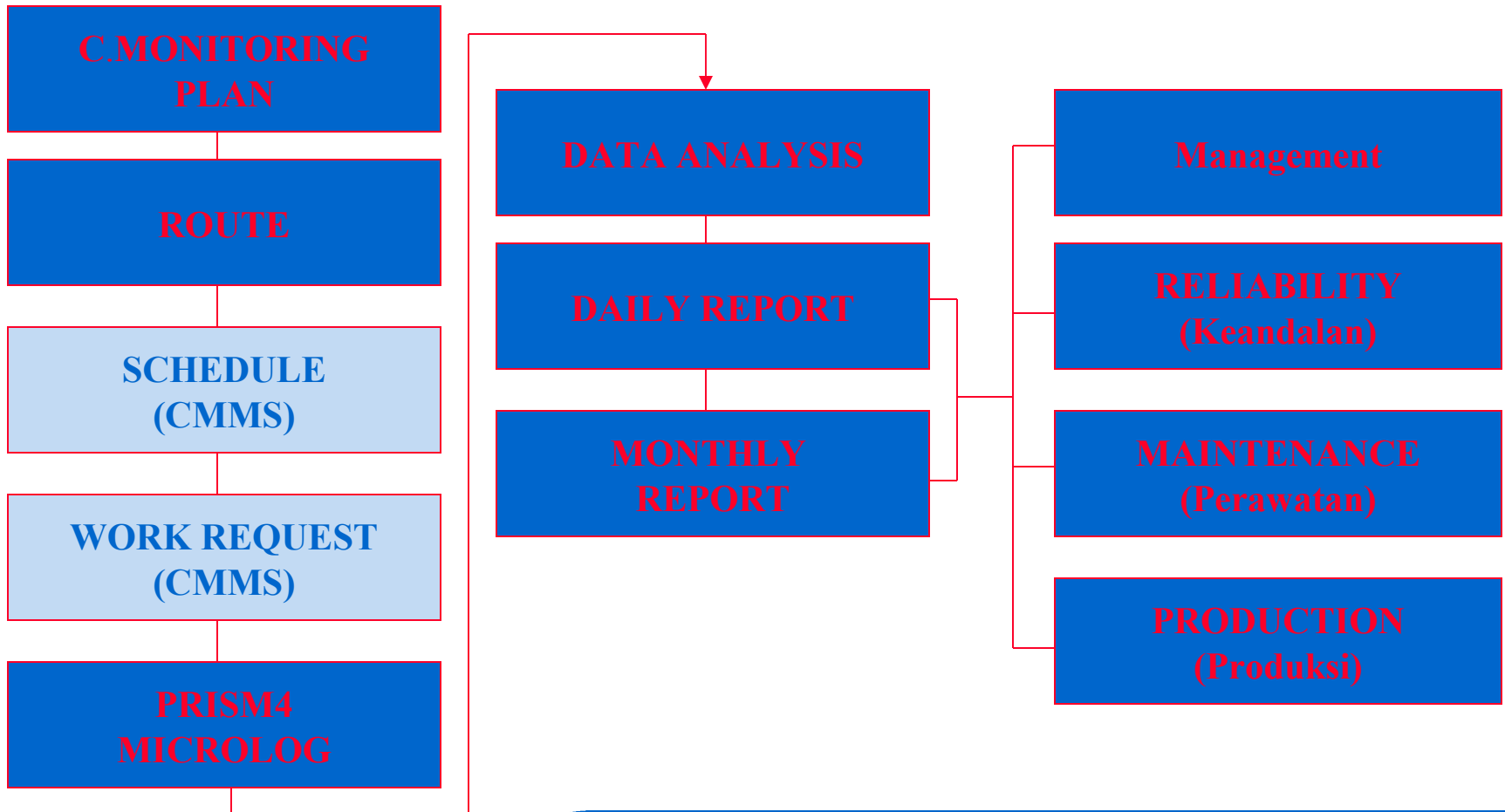
SKF

How Do Predictive Maintenance Programs Work?

The Predictive Maintenance Cycle



Work Flow

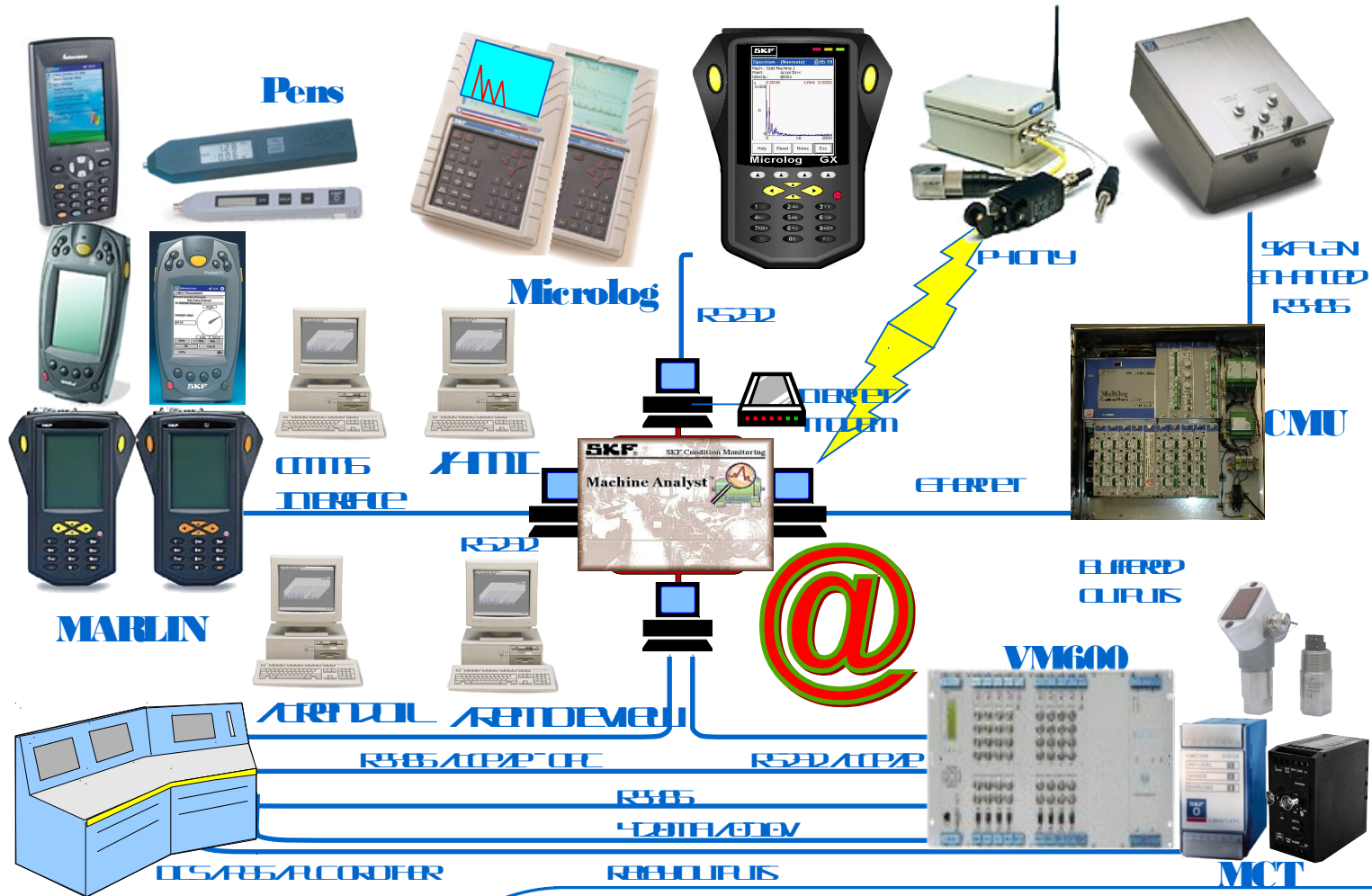


How are Measurements Performed?

... Sense and Convert Mechanical Energy Into An Electrical Signal



Our Integrated Platform for Condition Monitoring



Predictive Maintenance Measurements

Vibration

Temperature

Lube Oil

Process

-Pressure

-Flow

-Speed



PT. SKF RELIABILITY SYSTEMS INDONESIA

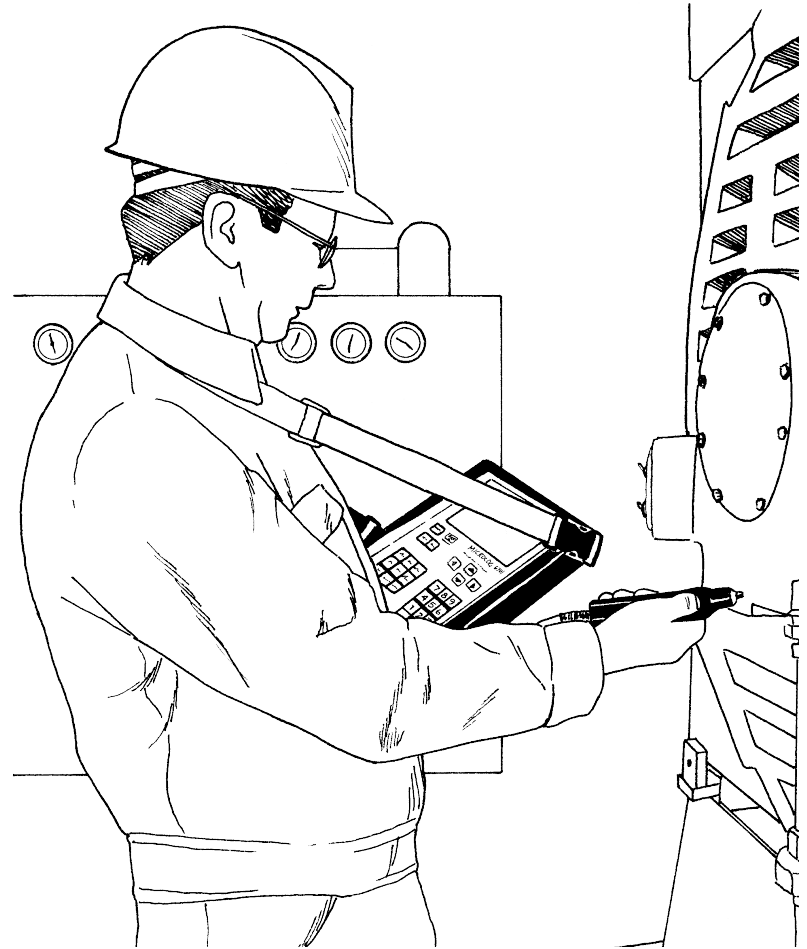
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3

Basic Vibration Philosophy

Measuring Vibration

Vibration is considered one of the most effective measurements for monitoring the condition of rotating machinery



SKF RELIABILITY SYSTEMS INDONESIA

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Measurable Vibration Characteristics

Frequency

- **What is vibrating?**

**Amplitude
vibrating?**

- **How much is it**

Phase Angle

- **How is it vibrating?**

Vibration Frequency Measurements & Units

- **RPM** (**R**evolutions **P**er **M**inute)
- **CPM** (**C**ycles **P**er **M**inute)
- **CPS/Hz** (**C**ycles **P**er **S**econd/**H**ertz)
- **Orders** of **Running Speed** (how many times event occurs per shaft revolution)

Vibration Amplitude Measurements & Units

Displacement (Distance)

Mils or Microns

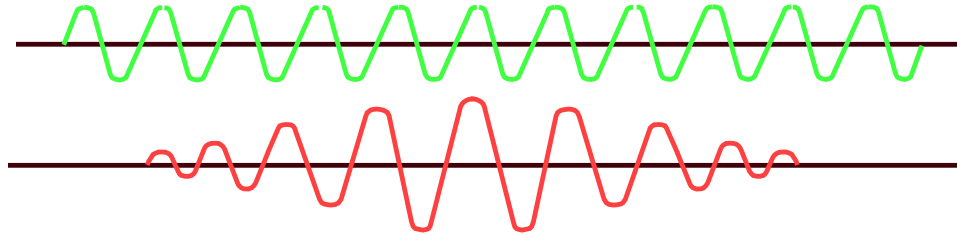
Velocity (Speed - Rate of change of displacement)

In./Sec. or mm/Sec.

Acceleration (Rate of change of velocity)

G's - In./Sec.² or Meter/Sec.²

Amplitude vs. Frequency



When analyzing vibration we look at two components of the vibration signal, its **amplitude** and its **frequency**.

- **Vibration amplitude** indicates the **severity** of the problem.
- **Vibration frequency** indicates the **source** of the problem.

Displacement Probe/Eddy Probe

- Measure **Relative Distance Between Two (2) Surfaces**
- **Accurate Low Frequency Response**
- **Limited High Frequency Sensitivity**
- **Require External Power Source**



Velocity Sensors



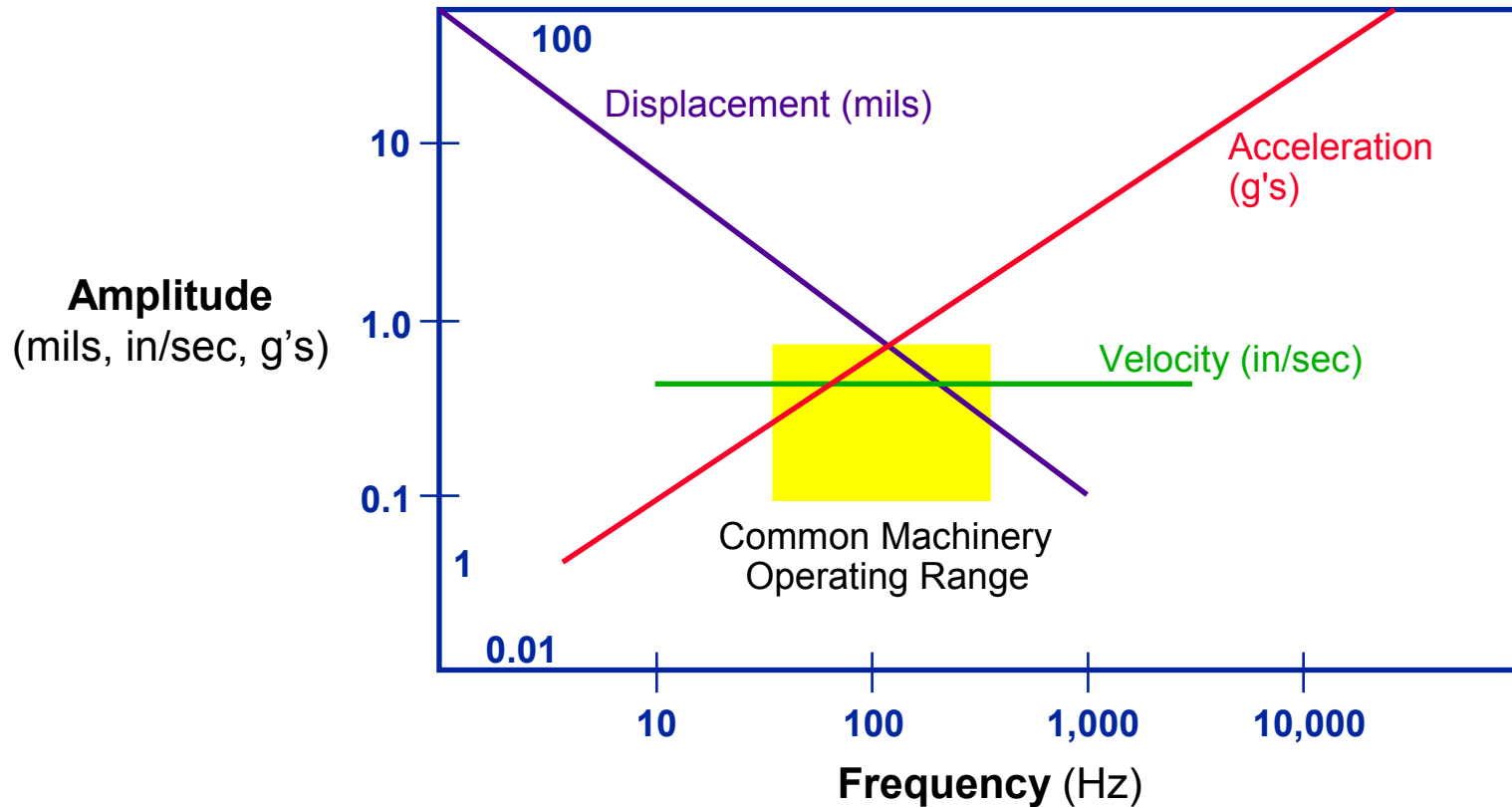
- **Often Measure Bearing Housings or Machinery Casing Vibration**
- **Effective in Low to Mid Frequency Range (10 Hz to around 1,500 Hz)**
- **Self Generating Devices**
- **Are Electro-Mechanical Devices With Moving Parts That Can Stick or Fail**

Accelerometers



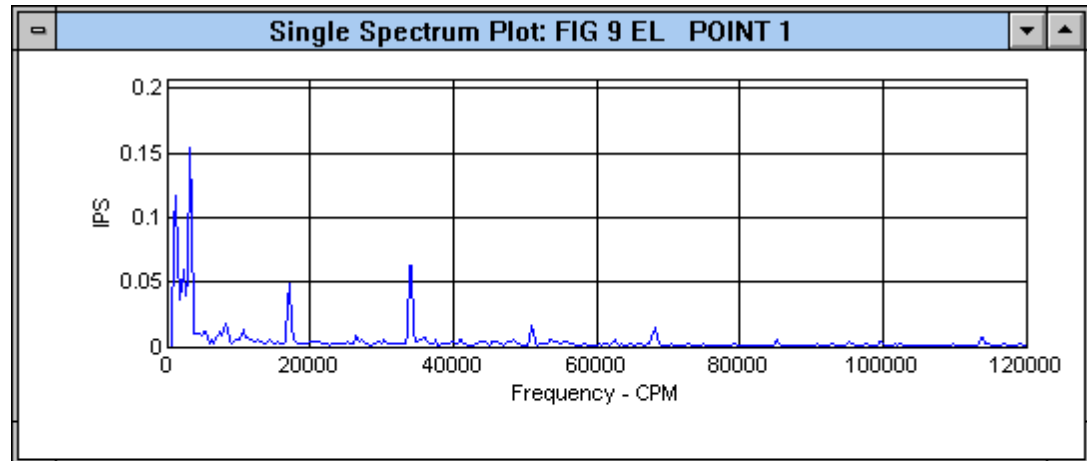
- **Rugged Devices**
- **Operate in Wide Frequency Range (Near 0 to above 400 kHz)**
- **Good High Frequency Response**
- **Some Models Suitable For High Temperature**
- **Require Additional Electronics**

Sensor Relationships

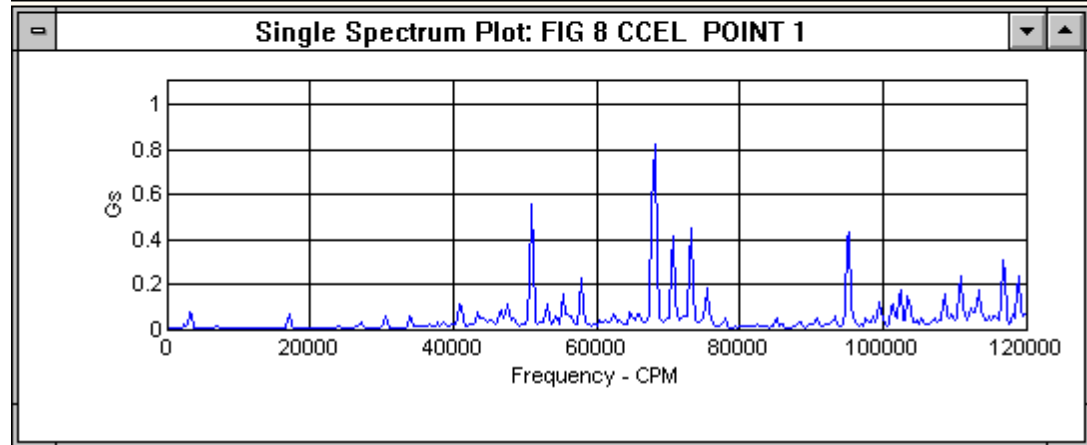


Same Data in Velocity and Acceleration

**Velocity
Spectrum**



**Acceleration
Spectrum**



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4

Vibration Analysis

Detection vs. Analysis

Detection

Alarm limits are established for each measurement. When the measurement's value exceeds its programmed alarm limits, the predictive maintenance software or data collector notifies the analyst of a problem.

Analysis

Once detected, analyzing exceptional measurements provides insight to the problem itself, and to its root cause.

Types of Alarms

- ☐ Overall Vibration Limits
- ☐ Spectral Enveloping
- ☐ Spectral Bands
- ☐ Phase Alarms

ISO Standard

Vibration Standard ISO 2372-1974

Vibration Severity	Velocity Range Limits and Machinery Classes ISO Standard 2372-1974			
	Small Machines	Medium Machines	Large Machines	
	Class I	Class II	Rigid Supports Class III	Flexible Supports Class IV
0.02				
0.03				
0.04				
0.06				
0.10				
0.16				
0.25				
0.39				
0.62				
1.00				
1.56				
2.50				
3.95				

Support classification according to ISO 2372

Flexible Support: The fundamental natural frequency of the machine/support system is lower than its main excitation frequency.

Rigid Support: The fundamental natural frequency of the machine/support system is higher than its main excitation frequency.

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SKF Condition Monitoring

P/N 31225300
Revision A (8-94)

SKF CONDITION MONITORING SYSTEM

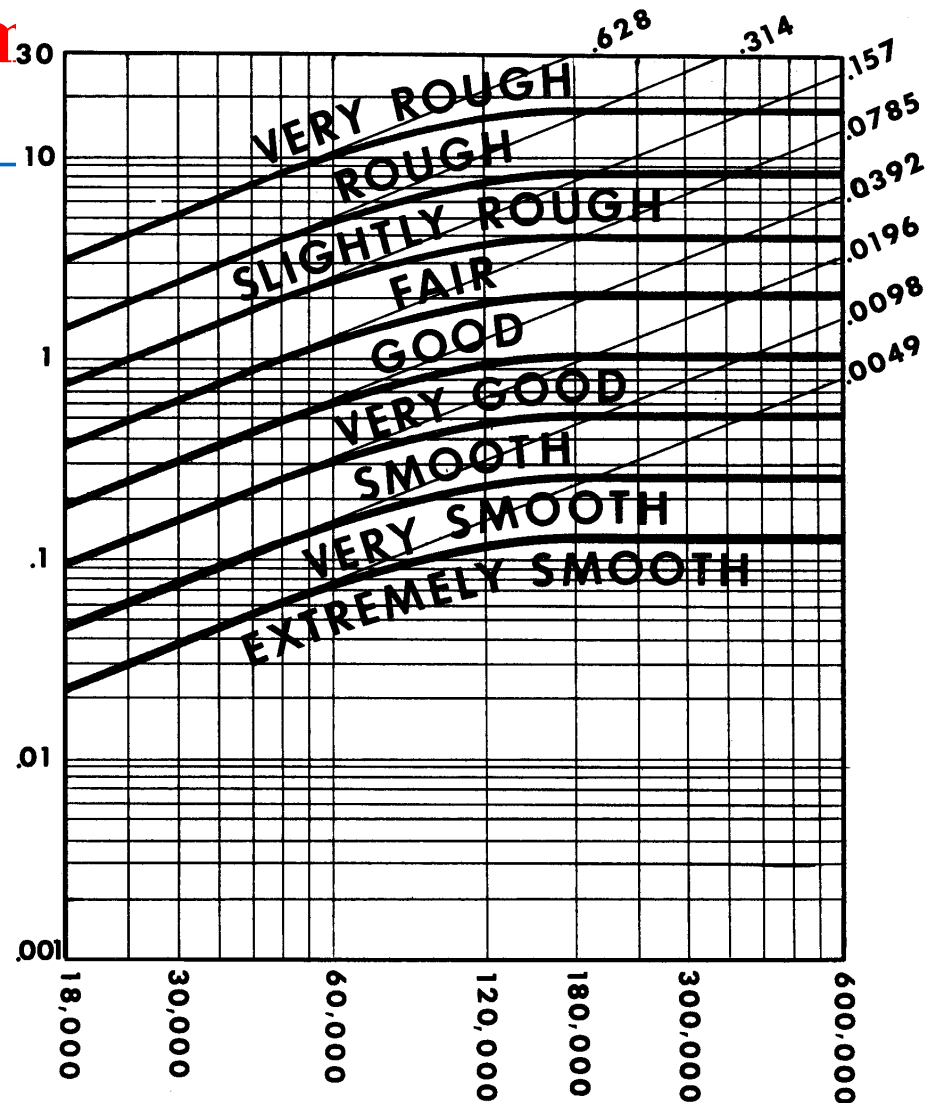
SKF

ISO Vibration Severity Chart

Velocity Severity		Velocity Range Limits and Machine Classes ISO Std.10816 - 1			
mm/s RMS	in/s peak	Small Machines Class I	Medium Machines Class II	Large Machines	
				Rigid Supports Class III	Flexible Supports Class IV
.36	0.02	Good	Good	Good	Good
.54	0.03				
.72	0.04	Satisfactory	Satisfactory	Satisfactory	Satisfactory
1.08	0.06				
1.80	0.10	Unsatisfactory (alert)	Unsatisfactory (alert)	Unsatisfactory (alert)	Unsatisfactory (alert)
2.87	0.16				
4.50	0.25	Unacceptable (danger)	Unacceptable (danger)	Unacceptable (danger)	Unacceptable (danger)
7.18	0.40				
11.14	0.62				
17.96	1.00				
28.00	1.56				
44.90	2.50				
70.94	3.95				

Overall Vib. Alarm

Assessing Overall Vibration Severity



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5

Vibration Spectrum

What is Vibration Analysis ?

Vibration Analysis is a measure of the energy being emitted from a surface due to forced and natural oscillatory movement. The energy is captured as vibrations and displayed in two common form:

- 1. Time vs Magnitude - known as time wave form**
- 2. Magnitude vs frequency - known as FFT**

Typical Vibration Analysis Display Formats

Overall Vibration (Trending)

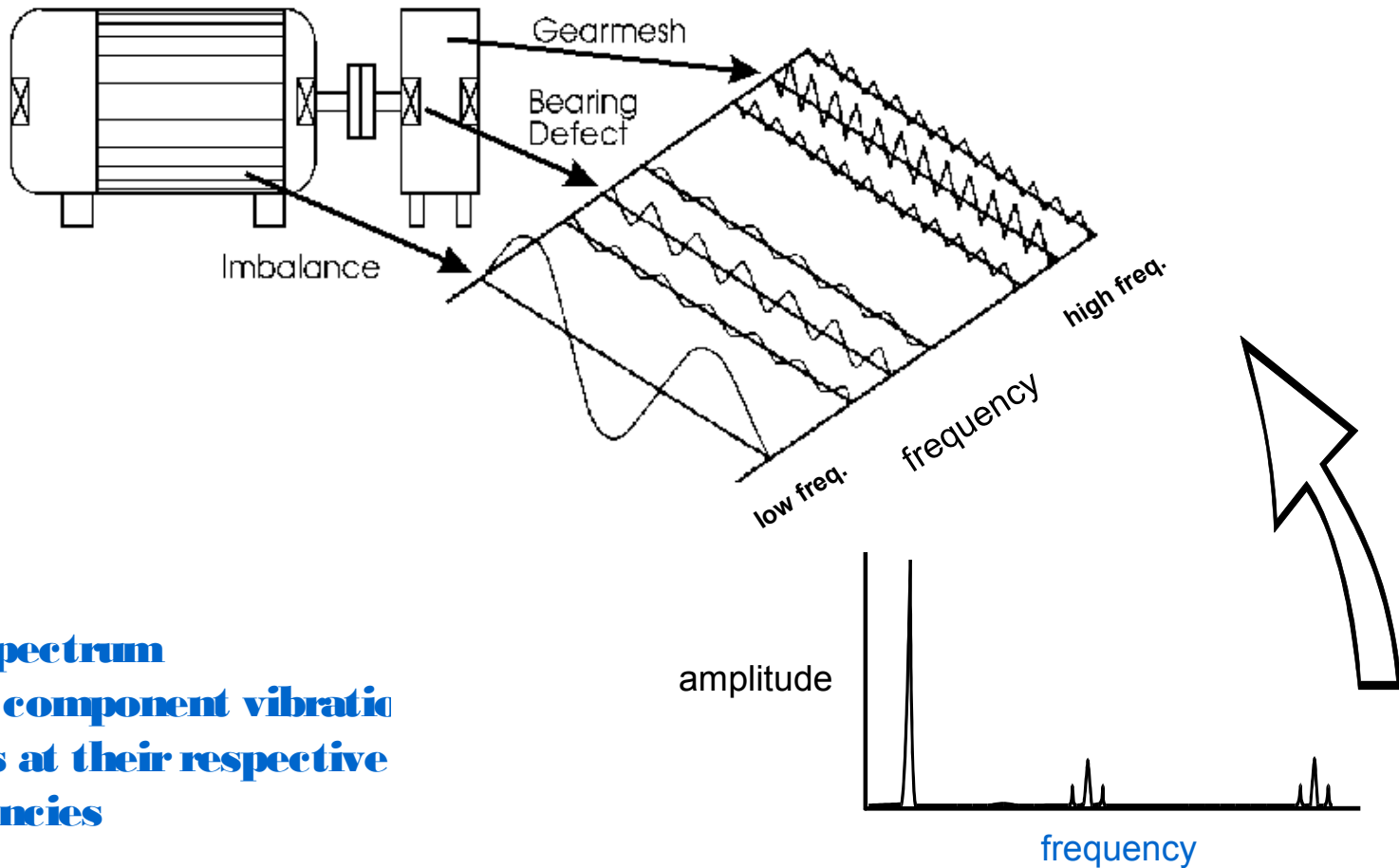
FFT Spectrum (Trending/Analysis)

Time Waveform (Trending/Analysis)

Other Processing Methods

- **Acceleration Enveloping**
- ***SE***

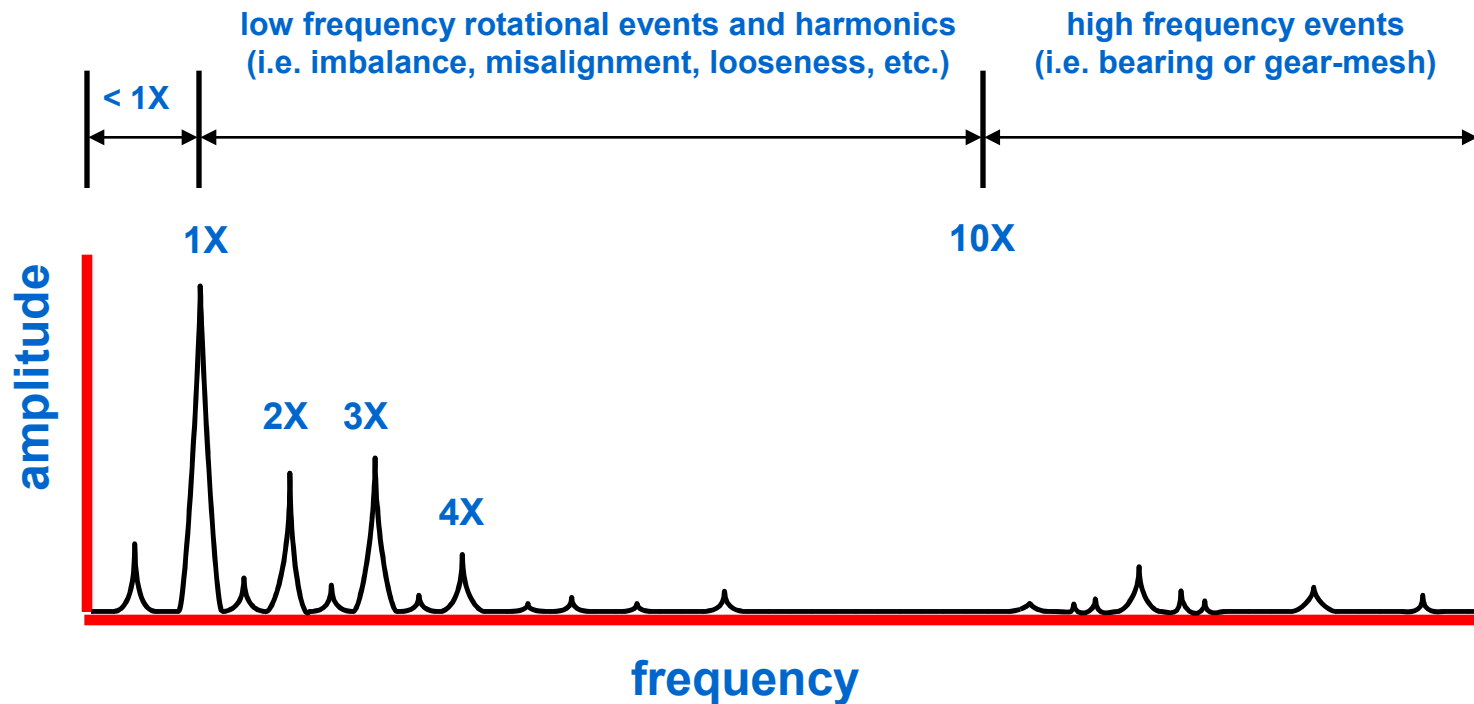
FFT Spectrum Analysis



FFT Spectrum
shows component vibratic
signals at their respective
frequencies

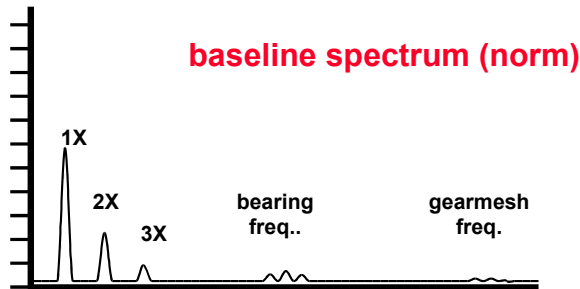
Spectral Pattern Recognition

(velocity spectrum)



Spectral Example

1



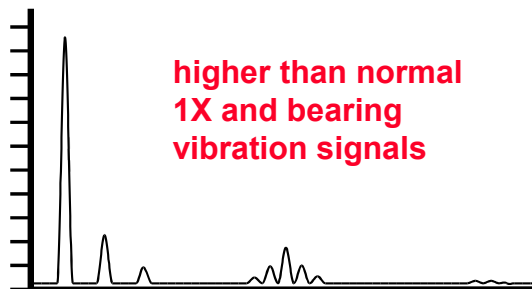
Known good measurement collected when machine was known to be in good running order.

2



1X running speed peak is significantly higher than normal. Indicates a vibration signal occurring once per shaft revolution. Typically caused by misalignment or imbalance.

3



Higher than normal 1X running speed peak and bearing fault frequency peaks indicate that 1X rotational problem has caused bearing damage.

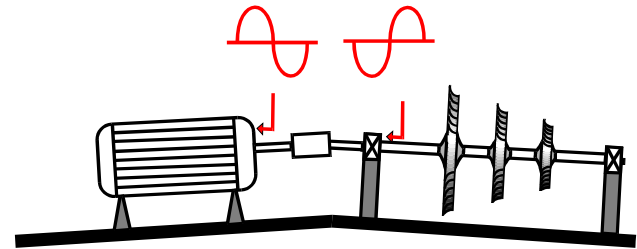
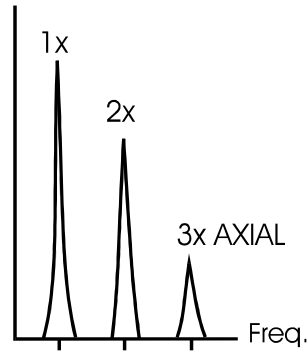
Sources of Vibration

Specific patterns in the spectral signatures are recognized as being related to specific machine problems.

- Misalignment 1x or 2x shaft speed
- Imbalance 1x
- Bent shaft 1x or 2x Axial
- Looseness 2x to 10x
- Gearmesh # of teeth x RPM
- Bearing Defects bearing defect frequencies

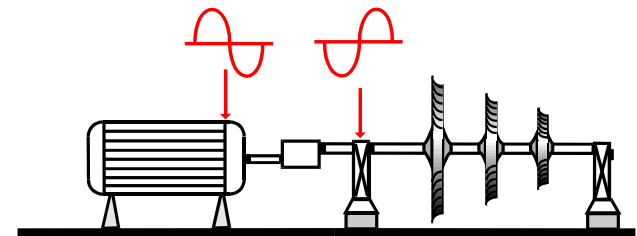
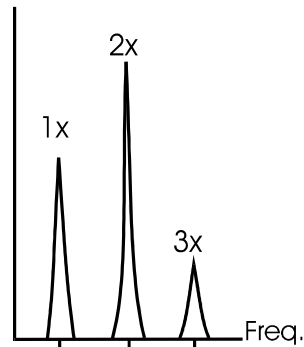
Misalignment

Angular Misalignment

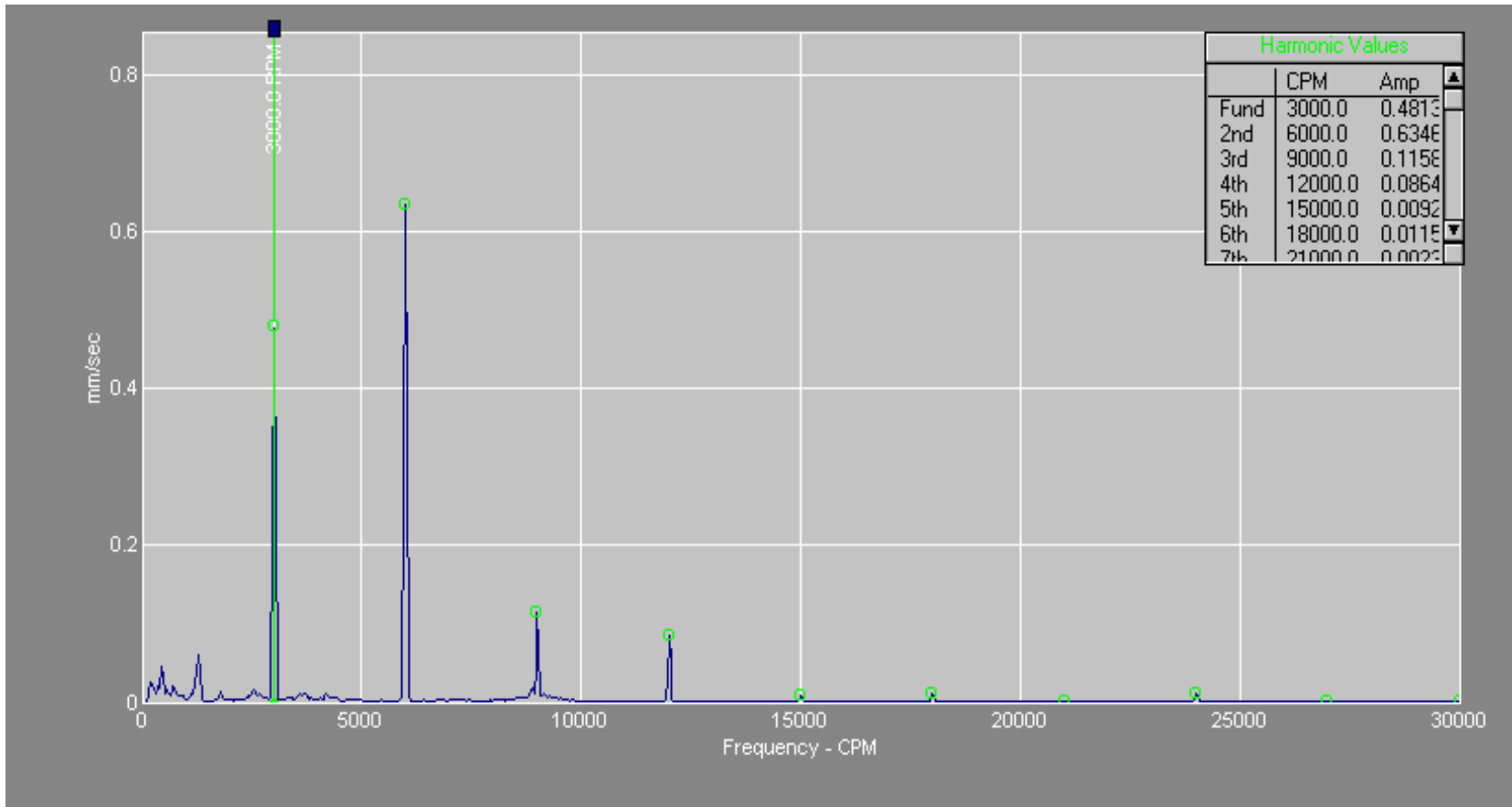


Parallel Misalignment

RADIAL

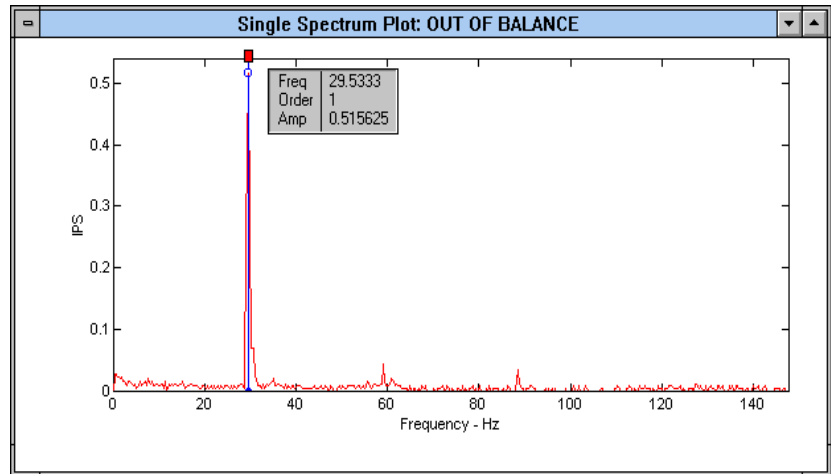


Velocity Measurement Showing Misalignment

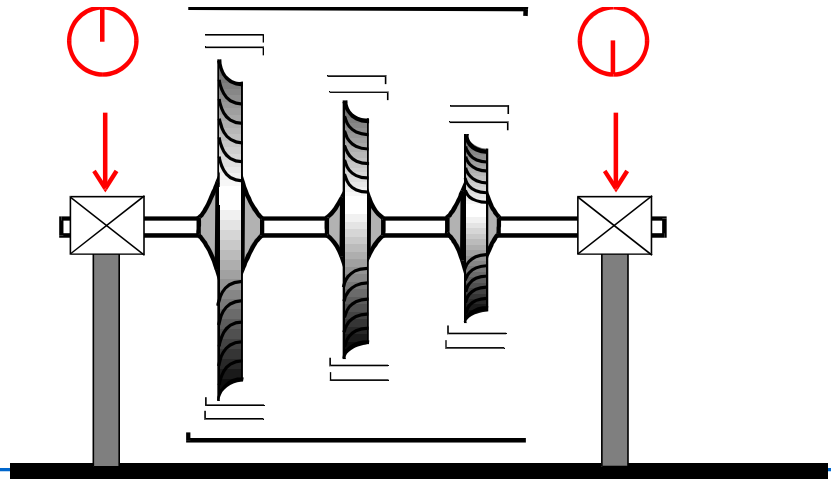


Imbalance - Diagnosis

static imbalance -
vibration across machine
“in phase”



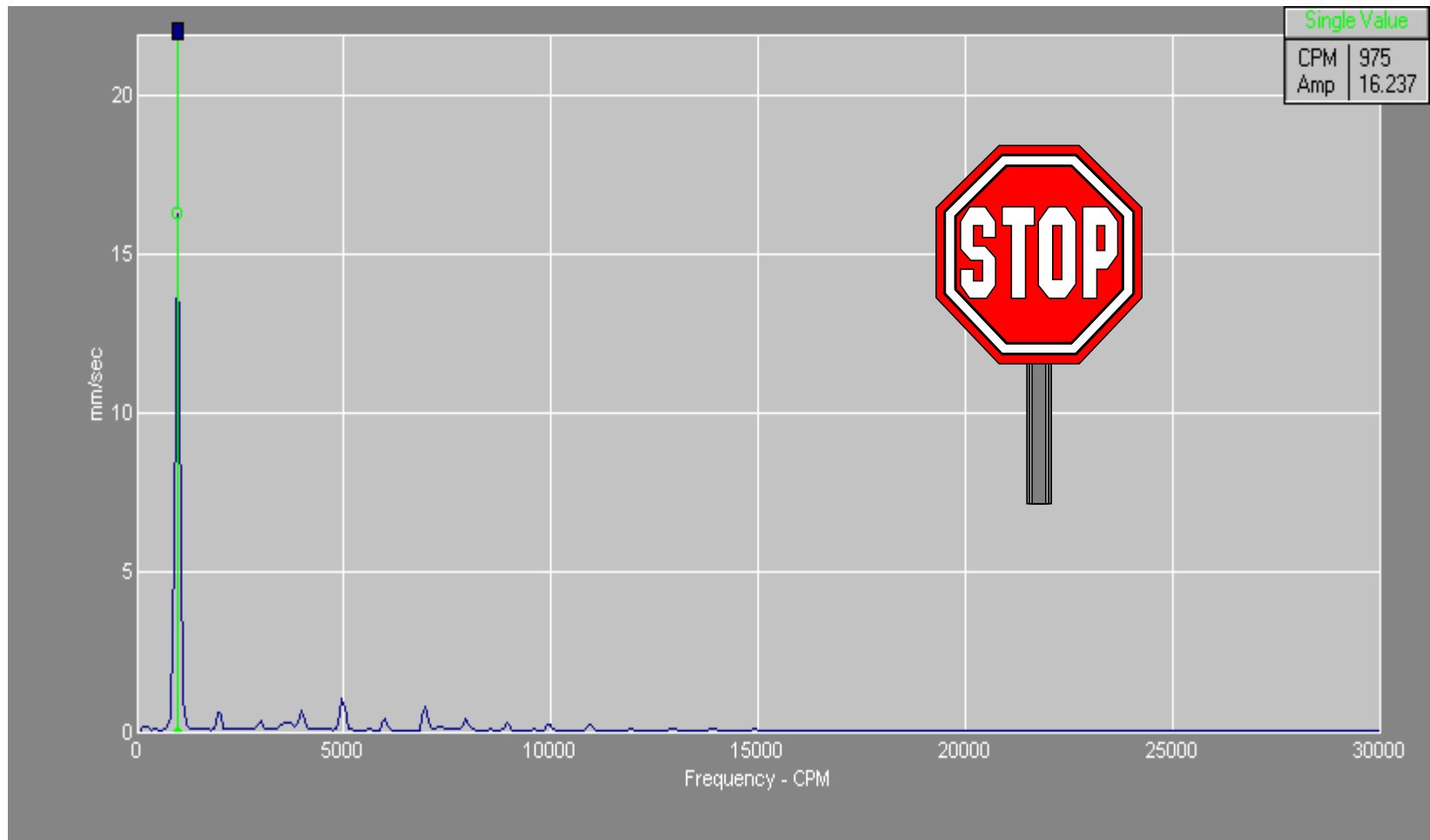
couple imbalance -
vibration across machine
“180 degrees out of phase”



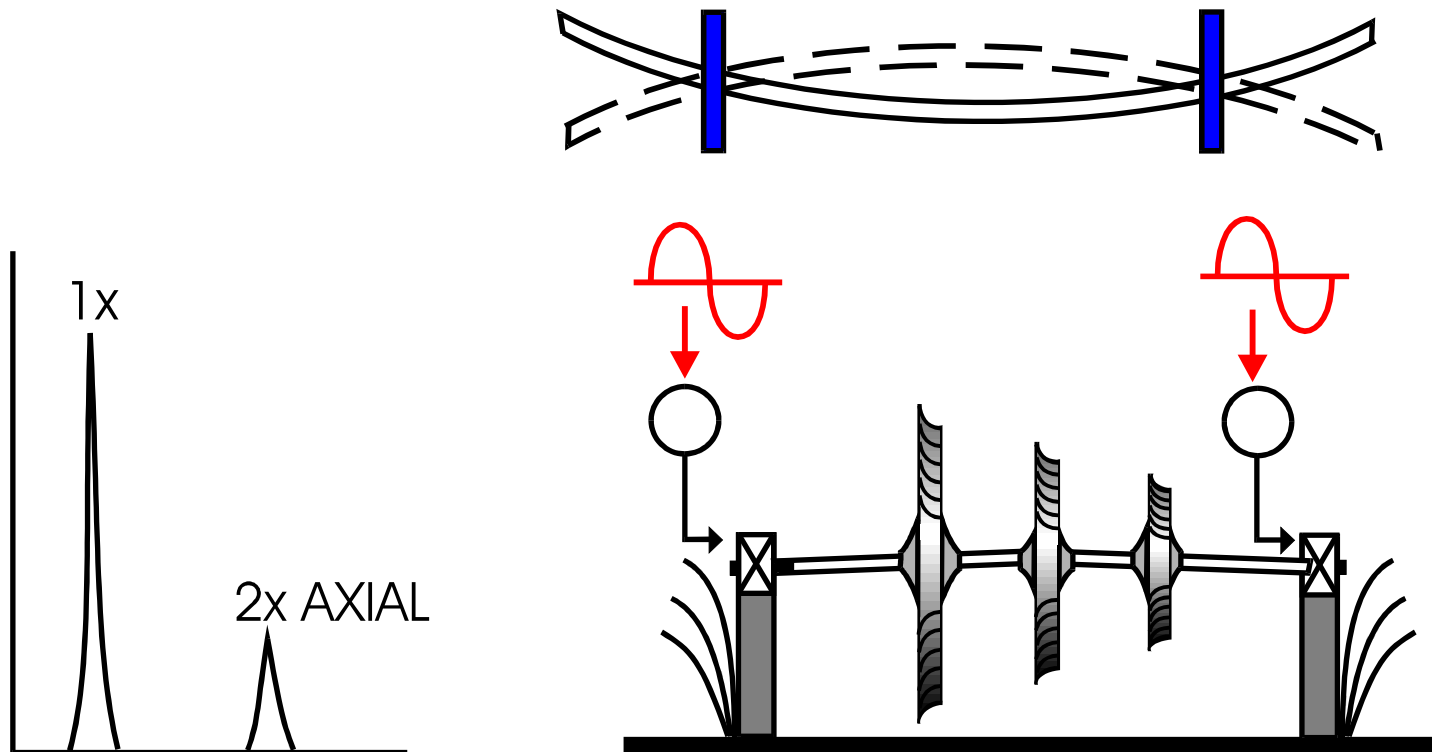
SKF RELIABILITY SYSTEMS INDONESIA

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Velocity measurement Showing Imbalance

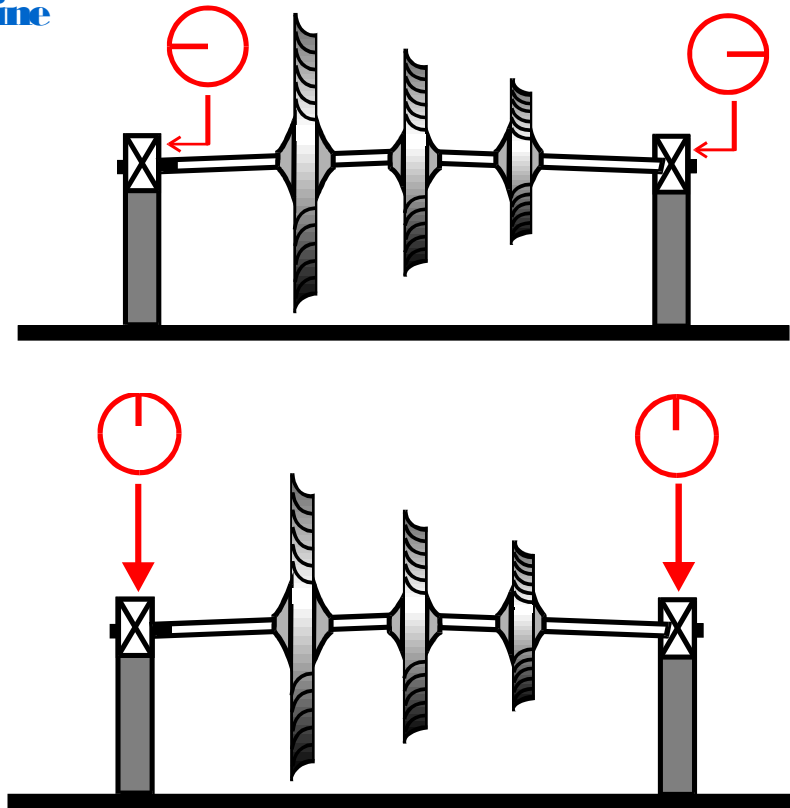


Bent Shaft



Bent Shaft - Diagnosis

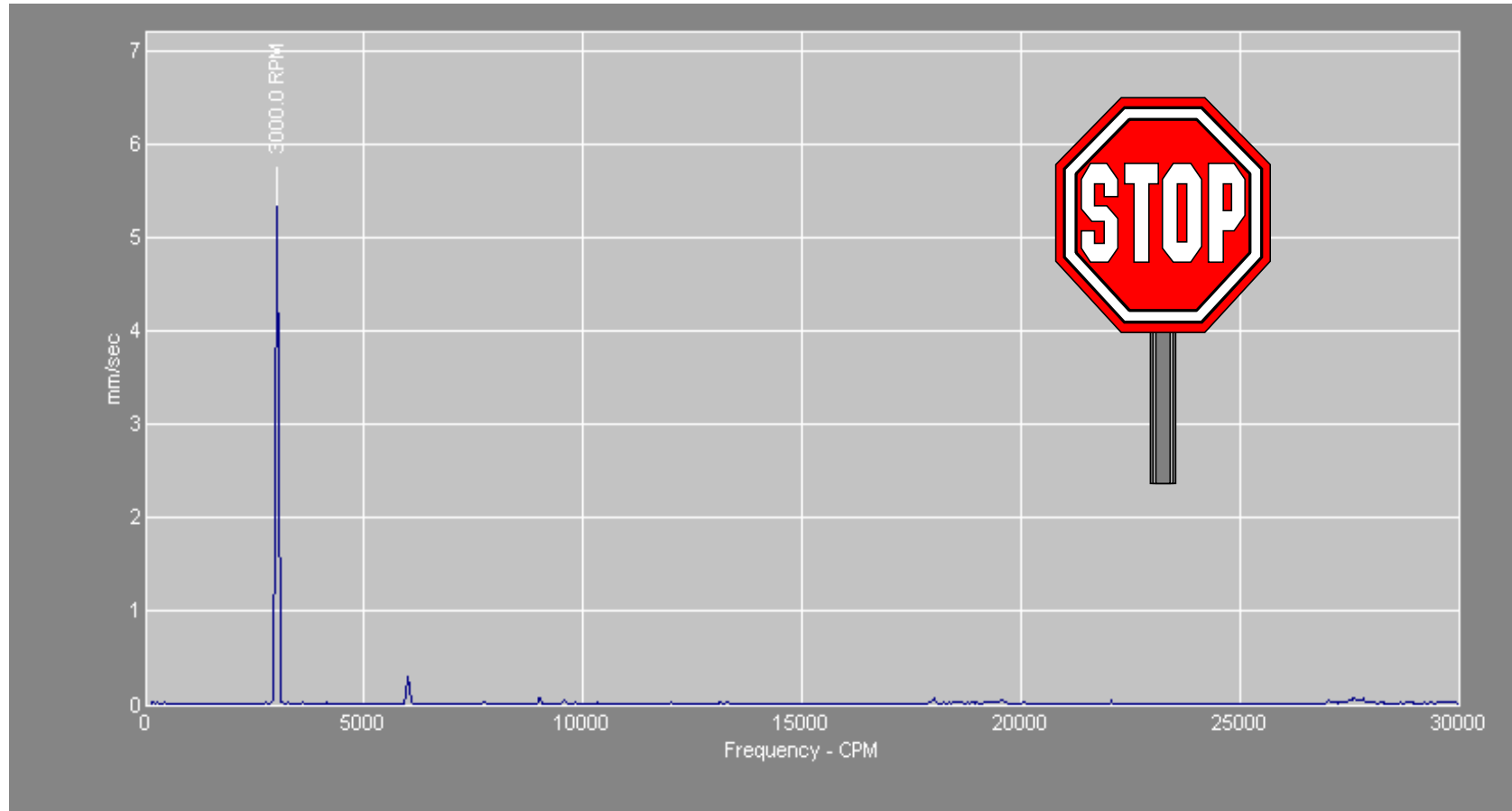
- 1 - axial measurements across machine typically 180 degrees out of phase, and ...**
- 2 - radial measurements across machine are typically in phase**



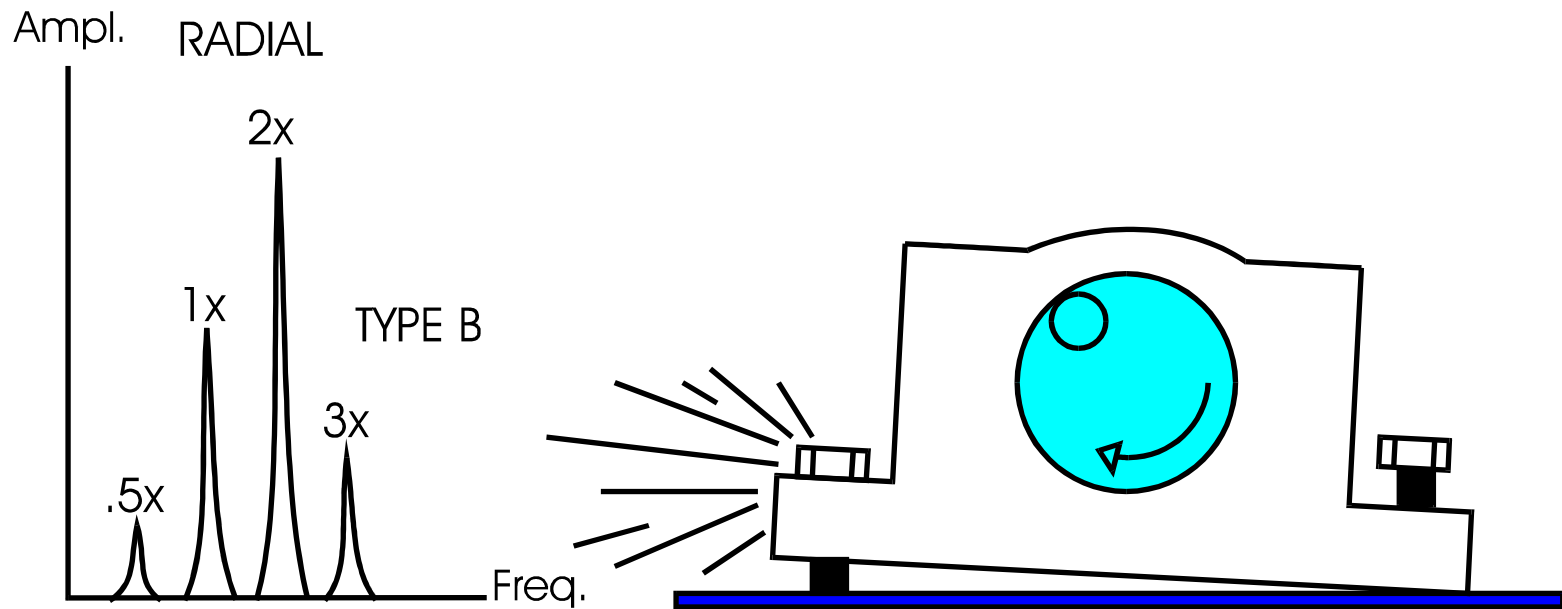
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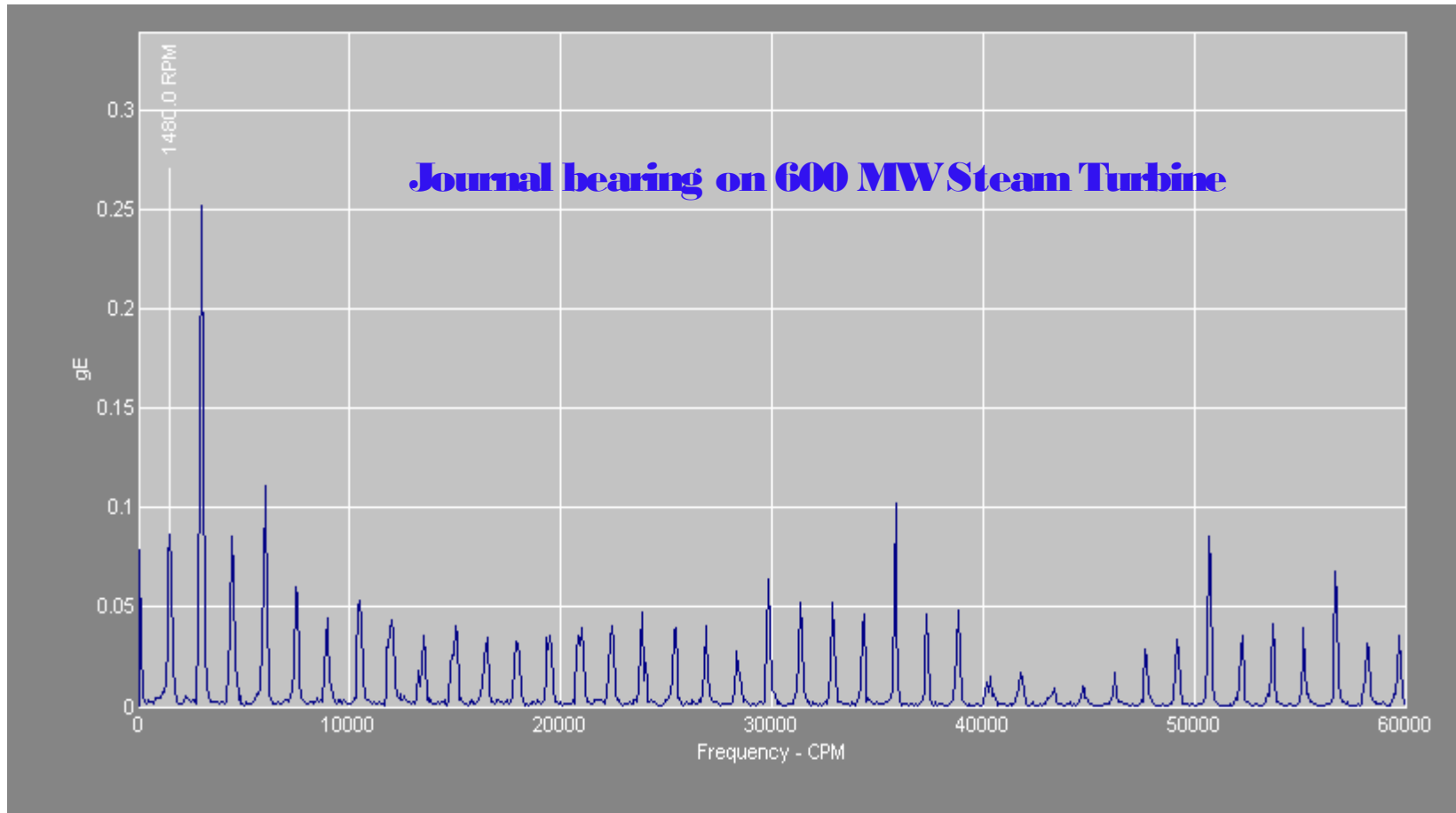
Velocity measurement Showing Bent Shaft Problem



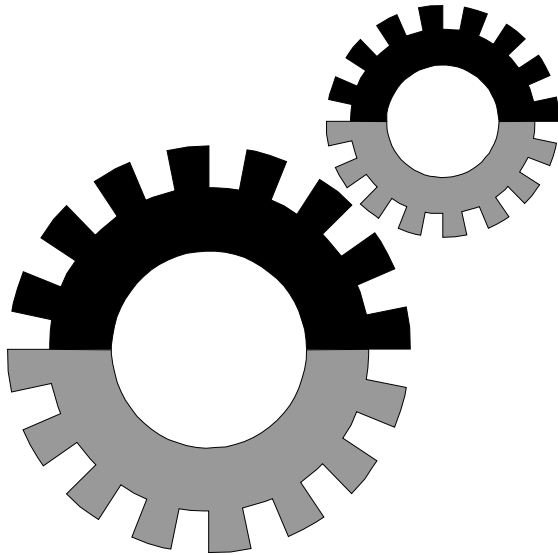
Mechanical Looseness



Enveloping measurement showing Mechanical Looseness Problem



Gear-mesh Frequencies

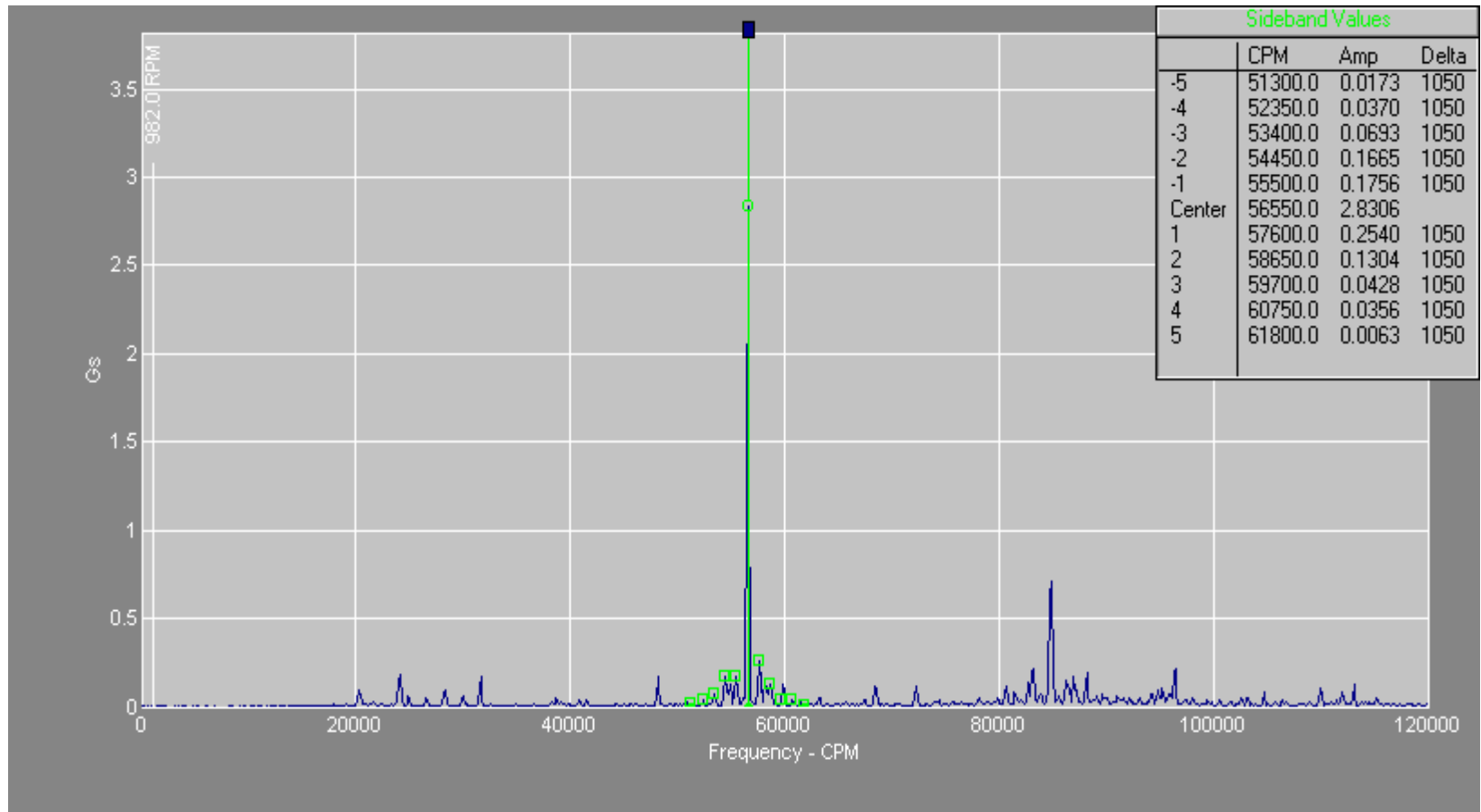


$$\text{GMF} = \# \text{ of teeth} \times \text{RPM}$$

Gear Defects

- The gear mesh frequency is equal to the gear shaft rotating speed times the number of teeth on that gear.
- The amplitude will vary with the load.
- Two equal gearboxes can have very different amplitudes under the same loads.
- **ALWAYS** collect a time domain spectrum on each gear shaft bearing.

Acceleration Measurement Showing Gear-mesh Frequencies

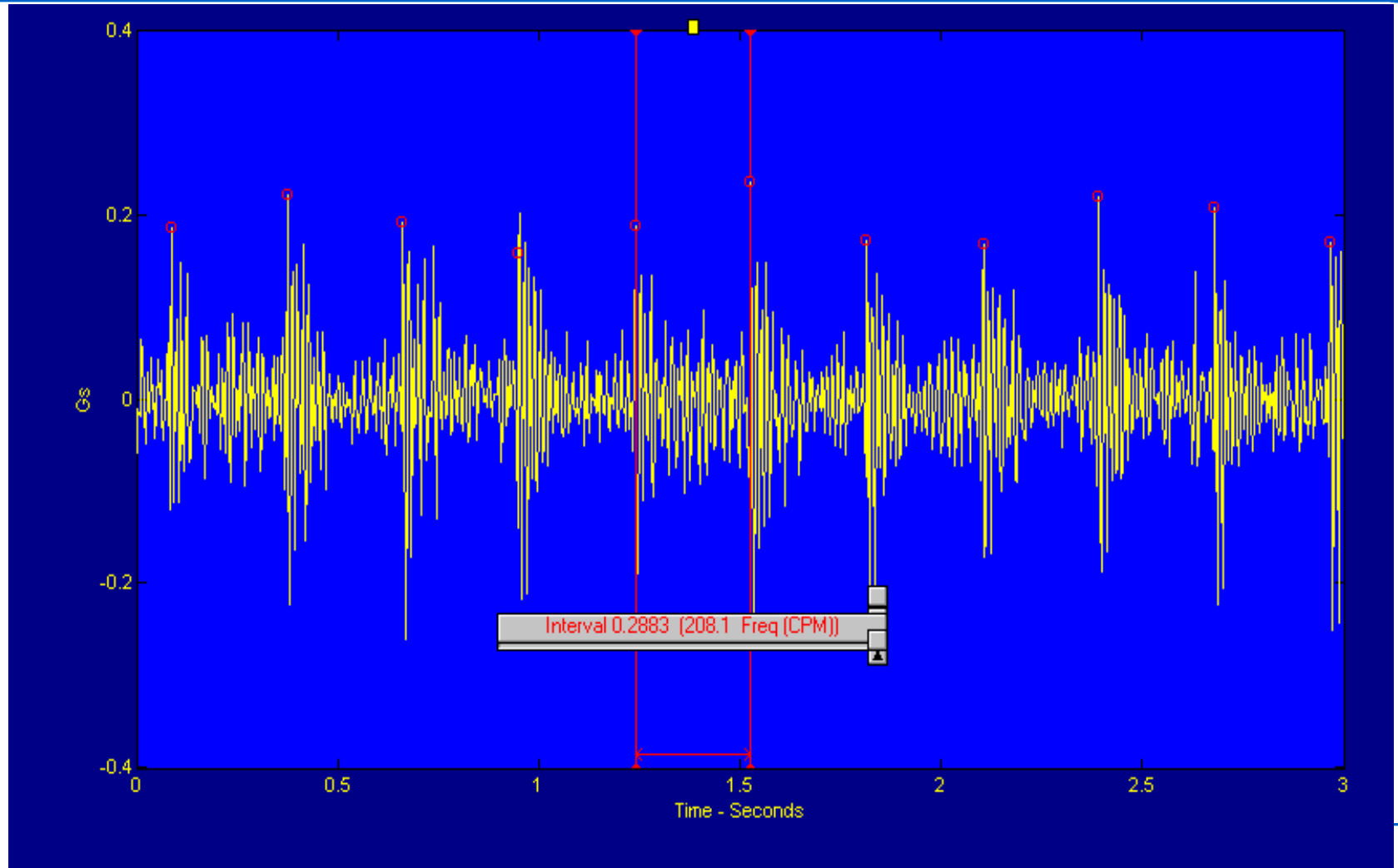


RELIABILITY SYSTEMS INDONESIA

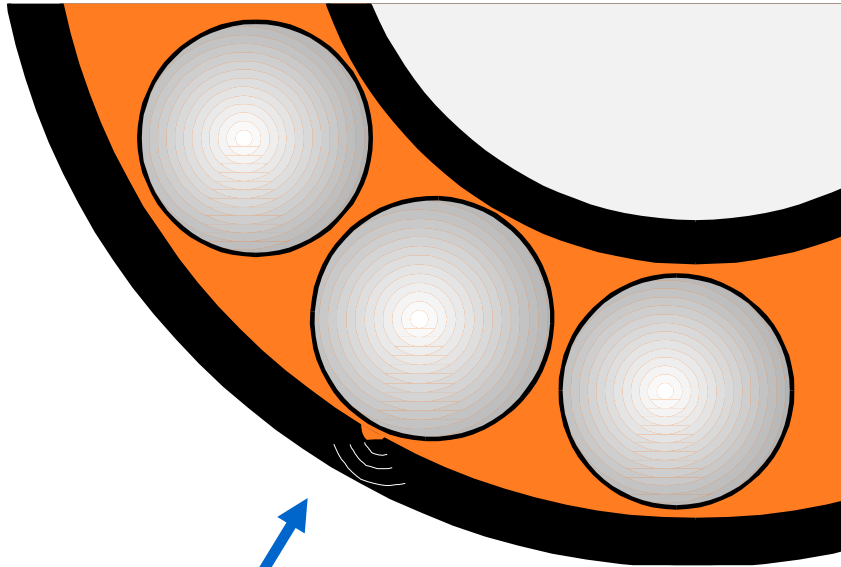
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Mining Gearbox Time Domain

208 RPM



Why Do Bearings Fail?



flaw on outer race

inadequate lubrication

- too much
- too little
- contaminated

excessive load caused by:

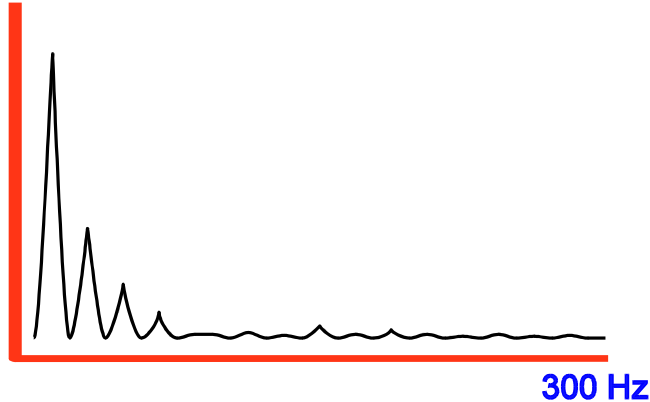
- misalignment
- imbalance
- bent shaft
- etc.....

improper handling or installation

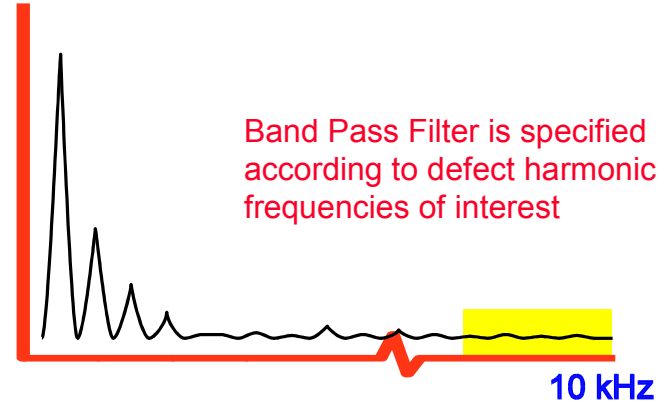
age

Enveloping

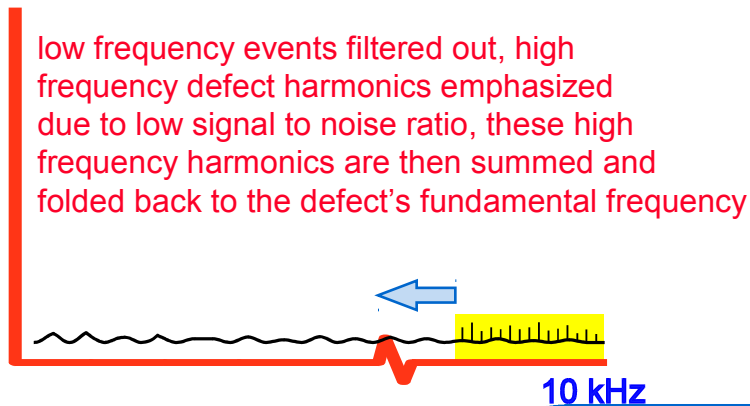
1 - Typical Velocity Spectrum



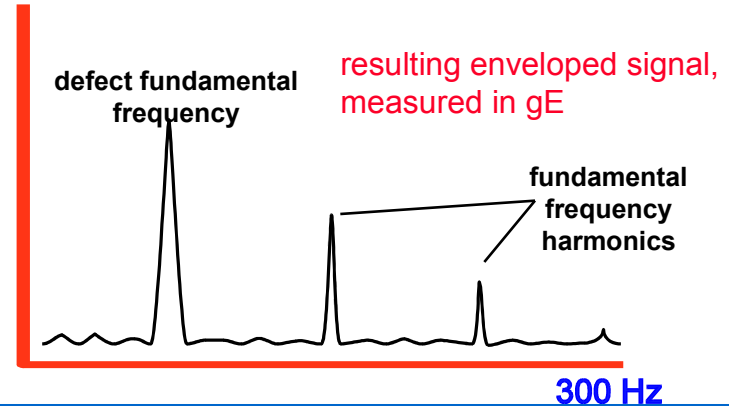
2 - Acceleration Enveloping Process



3 - Acceleration Enveloping Process



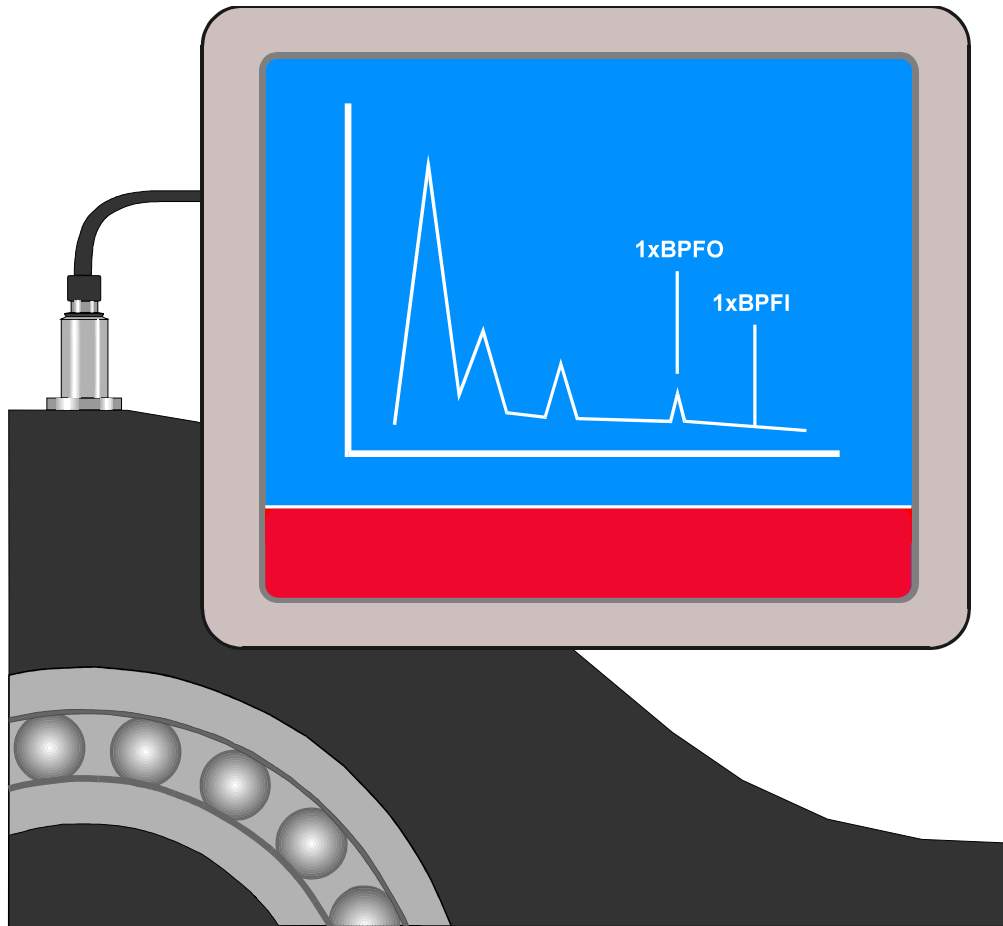
4 - Acceleration Enveloped Spectrum



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Bearing Defect Frequencies



BPFO

Ball Pass Frequency Outer Race

BPFI

Ball Pass Frequency Inner Race

BSF

Ball Spin Frequency

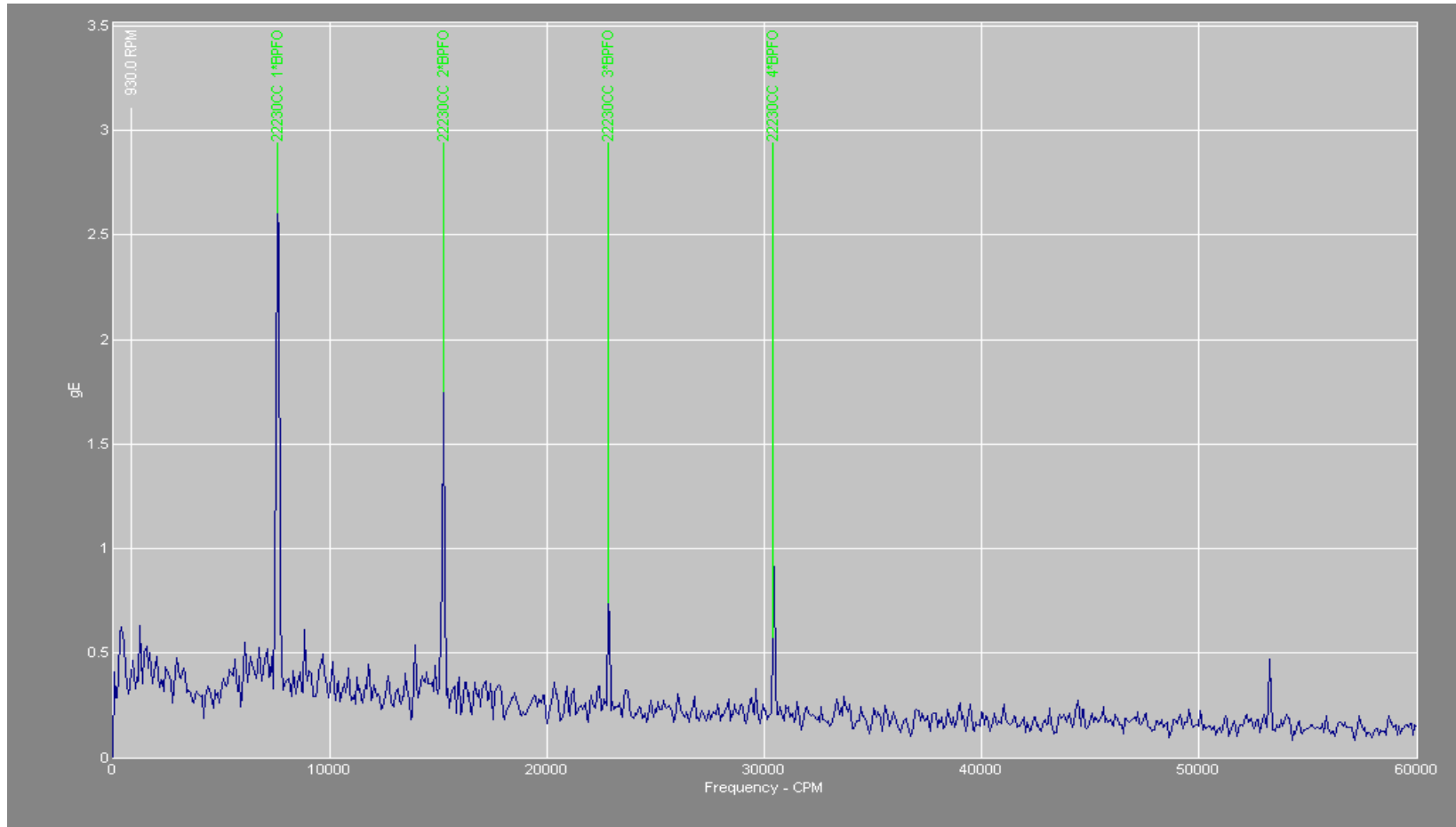
FTF

Cage Frequency or
Fundamental Train Frequency

BEARING DEFECT FREQUENCIES

SKF

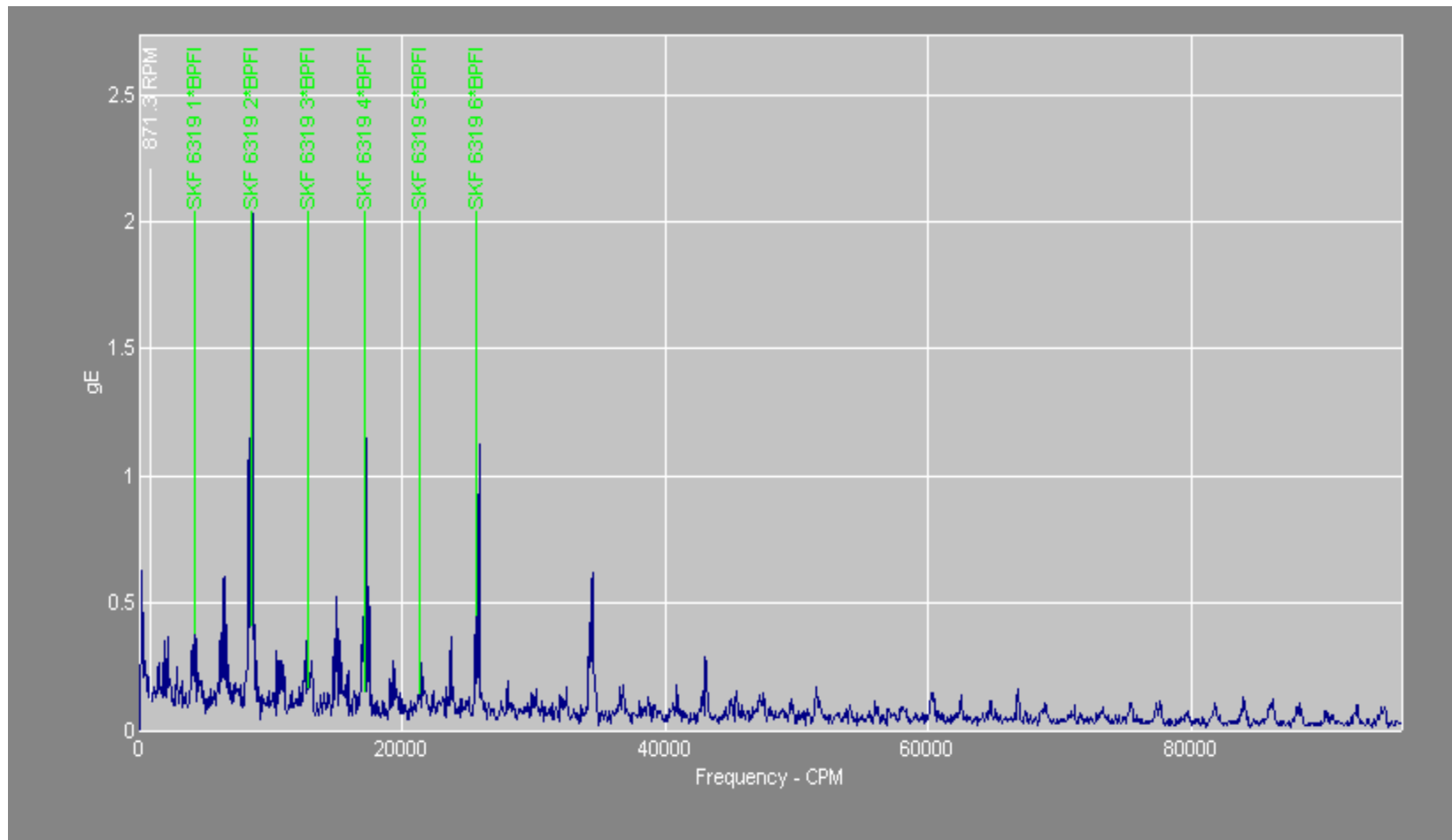
Acceleration Enveloping on bearing with Outer race defect



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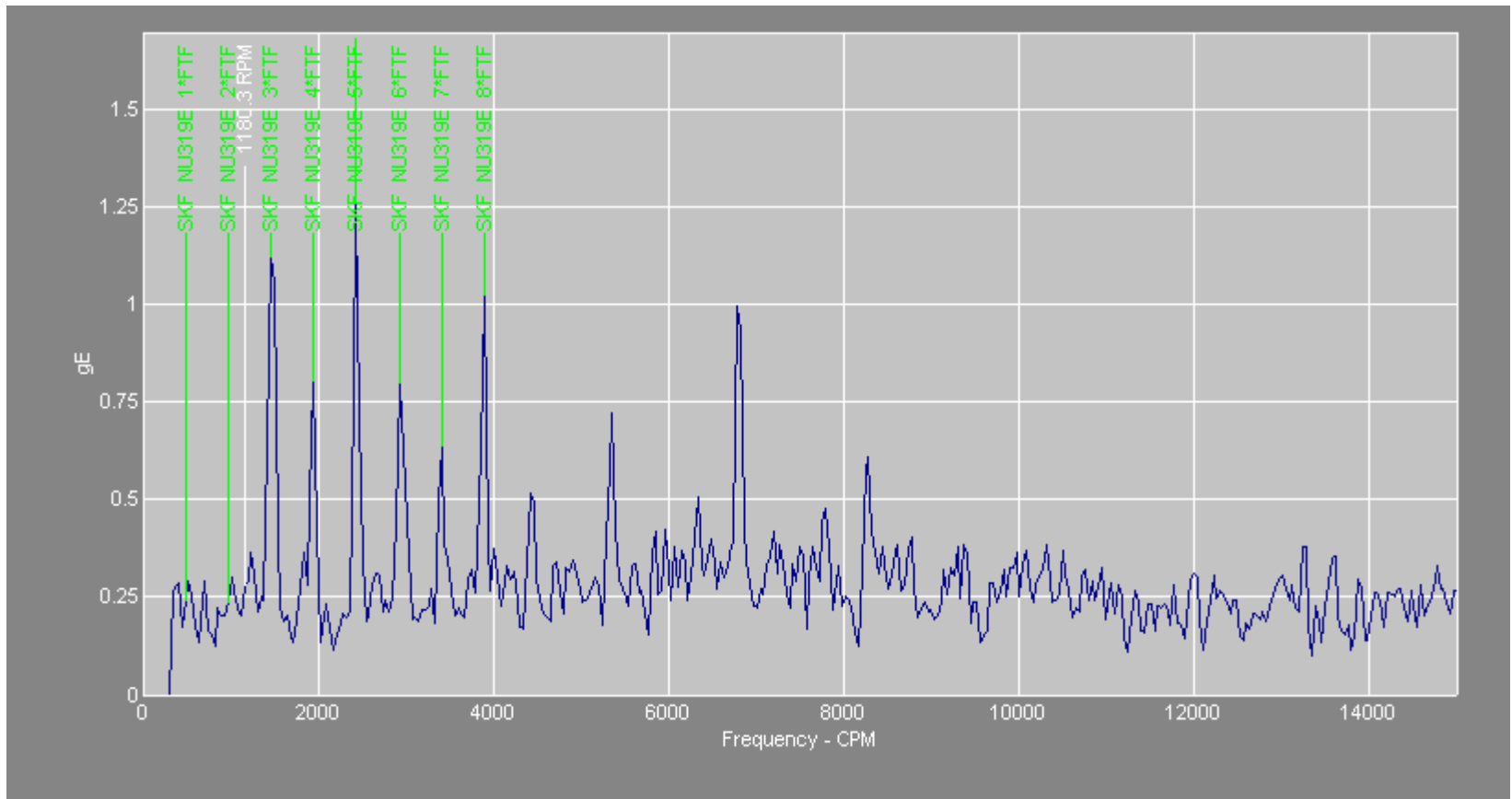
Acceleration Enveloping on bearing with Inner race defect



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Acceleration Enveloping on bearing with Cage defect



Enveloping Severity Chart

ISO 10816

Severity	Shaft Diameter Speed		
$gE_{peak\ to\ peak}$	Dia. between 200 & 500 mm and Speed < 500 RPM	Dia. between 50 & 300 mm & Speed between 500 & 1000 RPM	Dia between 20 & 150mm & Speed is either 1800 or 3600 RPM
0.1	Good	Good	Good
0.5	Satisfactory	Good	Good
0.75	Satisfactory	Satisfactory	Good
1	Unsatisfactory (alert)	Satisfactory	Satisfactory
2	Unacceptable (danger)	Unsatisfactory (alert)	Satisfactory
4	Unacceptable (danger)	Unacceptable (danger)	Unsatisfactory (alert)
10	Unacceptable (danger)	Unacceptable (danger)	Unacceptable (danger)

BERKUALITAS ISU TERBUKA

SKF

REQUEST FOR QUOTE

Thanks for Your Attention!!